

Divisive and Hierarchical Clustering Techniques to Analyse Variability of Rainfall Distribution Patterns in a Mediterranean Region

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Purpose of this document

This is an expanded literature-survey summary prepared for the project's mini-hackathon/review. It summarizes the published research rather than reproducing the copyrighted full paper.

1. Background and Problem

Mediterranean rainfall is highly irregular and can vary considerably from year to year. This variability makes it difficult to characterize rainfall behavior and determine whether the distribution of wet and dry periods is changing. The paper investigates whether clustering can reveal recurring rainfall distribution patterns over time.

2. Aim of the Study

The study aims to analyze rainfall distribution patterns and their variability over time. It uses a combination of divisive and hierarchical clustering techniques so that different clustering structures can be compared.

3. Dataset

The data consists of daily rainfall observations from 1889 to 1999 in the Alt Penedès region of northeastern Spain. The dataset was recorded at Vilafranca del Penedès and represents a very long historical rainfall record. The analysis considers annual rainfall as well as spring and autumn rainfall. Spring and autumn together represent, on average, about 70% of annual rainfall in the study region.

4. Methods

The paper uses K-means as a divisive clustering approach and also uses hierarchical methods, specifically average linkage and Ward's method. The purpose of applying more than one method is to see whether similar rainfall groupings are obtained and to understand which method better discriminates between patterns.

5. Results

The paper reports that K-means and Ward's hierarchical method produced similar classifications of the observations. The average-linkage method had less discrimination power. Even though individual techniques differed, the combination of methods was useful for identifying rainfall-distribution patterns and examining how those patterns changed over time.

6. Observed Pattern

One reported result is that patterns characterized by dry springs and wet autumns increased during the last decade of the study period. At the same time, the study did not find a consistent trend in annual rainfall. This illustrates how clustering can expose changes in the distribution of rainfall across seasons even when a simple annual total does not show a clear trend.

7. Significance

The paper is valuable for our project because it demonstrates a direct application of K-means and hierarchical clustering to an environmental variable. It also shows the benefit of comparing algorithms instead of assuming that one clustering method will always be superior.

8. Connection to Our Project

Our project is broader because we intend to analyze multiple climate variables together rather than focusing only on rainfall. However, the basic idea is the same: observations are grouped according to similarity and the resulting groups are interpreted as meaningful climate patterns. The paper supports our plan to compare K-means with Hierarchical Clustering and examine whether both methods identify similar structures.

9. Viva Points

If asked what this paper proves for our project, do not say it proves our algorithm will be correct. Instead say that it provides prior evidence that clustering methods can reveal useful rainfall and climate patterns, and that comparing K-means with hierarchical approaches can provide additional insight into the stability and interpretability of the groups.

Reference: See the publisher record/DOI for the original article.